

Event A: May 2024

This is a G5 storm event that occurred during a period of high solar activity in solar cycle 25. It was named the **Gannon (or Mother's Day) storm** and was the **biggest geomagnetic storm within the last 2 decades** (Bowler, 2025).

Seven CMEs were emitted between the 7th and 11th May 2024, and some of their speed exceeded 1000 km/s (Horne, 2026). The CMEs began hitting Earth's magnetic field from the **10th May** (Horne, 2026).

The aurora australis could be seen as far north as Uluru and Townsville, which is incredibly rare (Bowler, 2025). Spectacular aurora displays were seen in previously unheard-of latitudes like northern Italy (Flemming et al, 2024).

Impacts of the storm included:

- Geomagnetically induced currents (can cause power outages)
- Drop in satellite altitude (aka satellite drag)
- Satellite charging (can damage satellites and disrupt service)
- GPS & GNSS errors
- Degradation of HF communication (important for marine operations & aviation)
- Auroras at unusually low latitudes
- Plane diversions

(Horne et al, 2026; Bowler, 2025)

References:

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Flemming, T., Hewson, G. and Hindeleh, B. (2024). *Aurora borealis and australis dazzle viewers across the world three nights in a row*, ABC News, 13 May. Available at: <https://www.abc.net.au/news/2024-05-13/aurora-borealis-and-australis-captured-across-the-globe/103836752>

Horne, R.B., Angling, M.J., Attrill, G.D.R. et al. (2026). *The May 2024 geomagnetic storm: UK experience and perspective*, *Royal Society Open Science*, 13(4), 251943. Available at: <https://royalsocietypublishing.org/rsos/article/13/4/251943/481540/The-May-2024-geomagnetic-storm-UK-experience-and>

Event B: March 2015

Space Weather Events Storm Information

This is a G3/4 event that impacted earth on the 17th March 2015, was named the **St. Patrick's Day solar storm**. This was the **most significant geomagnetic storm of solar cycle 24** and was caused by CMEs ejected from the sun at around 02:00 UT on 15th March 2015 (NOAA SWPC, 2015). The initial speed of the CME was around 668 km/s (Wu et al., 2016). The storm hit Earth's magnetic field around 04:45 UT on 17th March, and it was fully recovered by the 25th (Murphy et al., 2018).

Impacts included spectacular aurora visible as far south as Colorado, severe ionospheric density depletion (will have disrupted satellite and radio communication), geomagnetically induced currents at power plants (NOAA SWPC, 2015) and strong GNSS/ GPS disturbances (Jacobsen and Andalsvik, 2016).

References:

- Jacobsen, K.S. and Andalsvik, Y.L. (2016). Overview of the 2015 St. Patrick's day storm and its consequences for RTK and PPP positioning in Norway. *Journal of Space Weather and Space Climate*, 6, A9. <https://doi.org/10.1051/swsc/2016004>
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- National Oceanic and Atmospheric Administration (NOAA), Space Weather Prediction Center (SWPC) (2015) *St. Patrick's Day geomagnetic storm*. Available at: https://www.swpc.noaa.gov/sites/default/files/images/u33/StPatrick%27sDay_Geomagnetic_Storm.pdf
- Wu, C.-C., Liou, K., Lepping, R.P., Hutting, L., Plunkett, S., Howard, R.A. and Socker, D. (2016). The first super geomagnetic storm of solar cycle 24: "The St. Patrick's Day event (17 March 2015)", *Earth, Planets and Space*, 68(151). Available at: <https://doi.org/10.1186/s40623-016-0525-y>

Event C: Nov 2019

These two weeks in November 2019 were geomagnetically quiet and there were **no geomagnetic storms** recorded ($K_p < 5$) (Matza et al., 2021).

This is because November 2019 was the **solar minimum** of solar cycle 24, where the sun's activity was calm right before the transition to solar cycle 25 in December 2019 (NASA, 2020).

References:

- NASA (2020). *Solar Cycle 25 Is Here*. NASA, NOAA Scientists Explain What This Means, NASA Science / NASA Scientific Visualization Studio. Available at: <https://svs.gsfc.nasa.gov/13714/>

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